Cambridge
O Level

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Cambridge Ordinary Level

ENVIRONMENTAL MANAGEMENT **5014/21**

Paper 2 **October/November 2017**

MARK SCHEME

Maximum Mark: 60

Published

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Question	Answer	Marks
1(a)(i)	1 677 000;	**1**
1(a)(ii)	25.8(%);; *(if answer incorrect, allow one mark for 1 677 000 ÷ 6.5 [1])*;	**2**
1(a)(iii)	Nashville, Memphis, Knoxville, Chattanooga; cities correctly paired with their population (659 000, 656 000, 185 000, 177 000);	**2**
1(b)(i)	*any two from:* soil type; pH; planting density / number of plants / eq; size of plots; species / variety / type / strain / breed of maize; fertilizer; pesticide; water (schedule) / eq; AVP;	**2**
1(b)(ii)	as a control experiment; to compare (with the other treatments);	**2**
1(b)(iii)	*any three from:* *plot A:* plants / crops are shorter / smaller; shorter / smaller cobs; lower yield; *differences calculated:* 26 cm difference; 0.6 cm difference; 0.3 tonnes difference;	**3**
1(b)(iv)	6.8(%);; *(if answer incorrect, allow one mark for* 6.3 – 5.9 = 0.4 (/ 5.9) *[1])*;	**2**

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Question	Answer	Marks
1(b)(v)	*any two from:* repeat the trial; use, more plots / more samples; on different farms; use other varieties of maize; measure plants again;	**2**
1(c)(i)	*any two from:* sample too small / should look at more plants; in only one part of the field / eq; yellow spots could be caused by other things;	**2**
1(c)(ii)	*any four from:* *ref to* a systematic or random method; transect laid out / eq; stated sample points; number of samples; random method, e.g. grid co-ordinates; use of, random tables / random number generator / eq; use of quadrats; size of quadrat; further detail, e.g. sample within the quadrat;	**4**
1(c)(iii)	*any two from:* saves time; saves fuel; less wheelings / eq; AVP;	**2**

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Question	Answer	Marks
2(a)(i)	*any two from:* switchgrass, is a renewable crop / can be replanted / eq; does not need (added) zinc; idea of carbon neutral / carbon neutral described / little contribution to, global warming / greenhouse gases;	**2**
2(a)(ii)	*any three from:* transport costs low; labour costs low; more carbon neutral due to shorter distance / less air pollution / eq; (transport is) not time consuming / easier / eq;	**3**
2(a)(iii)	*any three from:* less food for animals; so less meat production; for humans / humans eat maize / eq; switchgrass grows well in, poor / zn deficient soils / ORA; ethanol can be made from, other crops / wastes / eq; maize can be exported;	**3**
2(b)(i)	198 AND 91;	**1**
2(b)(ii)	*any two from:* higher costs of production / profit per dollar invested only slightly more; calculations to show this, e.g. switchgrass is 25 cents on the dollar and hay is 20 cents on the dollar; hay may be more use to them / switchgrass not a fodder crop / eq; AVP;	**2**

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Question	Answer	Marks
2(c)(i)	*one mark for a valid method and one mark for further detail:* systematic; e.g. from a list of all farms choose every e.g. tenth; random; e.g. from a list of all farms choose sample using random number generator; stratified / quota; e.g. choose a sample based on the, size / type of farm / choose number of farms of each type according to the number of each needed; self-selection / volunteer; e.g. advertise for farms to volunteer for the survey;	**2**
2(c)(ii)	*any two from:* much quicker; reduces, travel costs / cost of carrying out survey; AVP, e.g. a different answer might be given over the phone;	**2**
2(c)(iii)	subsidies / tax break / provide seeds / eq;	**1**
2(c)(iv)	*any three from:* biofuel comes from the photosynthesis of plants / plants take in CO_2 / eq; so carbon released is the same as carbon captured / carbon neutral; less fossil fuels are, used / burnt; a reduction in the additional carbon dioxide added to the atmosphere / less greenhouse gases released;	**3**

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Question	Answer	Marks
3(a)	*any three from:* bacteria fix nitrogen; from the air; decomposition (of dead plant material / organic matter); humus added to the soil; nutrient cycling; better soil structure / drainage;	**3**
3(b)(i)	orientation with linear scale; *axes labelled:* average length of maize roots / mm; zinc concentration / ppm; plots correct;	**4**
3(b)(ii)	negative correlation / as Zn concentration increases root length decreases;	**1**
3(b)(iii)	28(mm); shown on graph;	**2**
3(b)(iv)	*any three from:* long roots absorb, more minerals / nutrients from deeper down; (long roots) so more growth / yield; also (absorb) more water; so less likely to die in hot weather; crops less likely to fall over / more resistant to wind; bind soils better / less soil erosion;	**3**

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Question	Answer	Marks
3(c)	*any four from:* remove, waste tips / chemical waste; fill holes with suitable material, e.g. infill / landfill / mining / seal the mine; waste / overburden / water; add topsoil; add fertiliser; cover with, trees / grass / plant species; grow plants that absorb toxic metals (and remove the plants); landscaping / reprofiling; to reduce drainage of toxic substances into water courses;	**4**

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